

11. Perform Affine Transformation on the image.

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Aim:

To perform affine transformation on an image using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2
```

```
import numpy as np
```

```
img=cv2.imread(r"C:\Users\Maruthi\Downloads\METAL.jpg")
```

```
rows,cols=img.shape[:2]
```

```
pts1=np.float32([[50,50],[200,50],[50,200]])
```

```
pts2=np.float32([[10,100],[200,50],[100,250]])
```

```
M=cv2.getAffineTransform(pts1,pts2)
```

```
dst=cv2.warpAffine(img,M,(cols,rows))
```

```
cv2.imshow("Original",img)
```

```
cv2.imshow("Affine",dst)
```

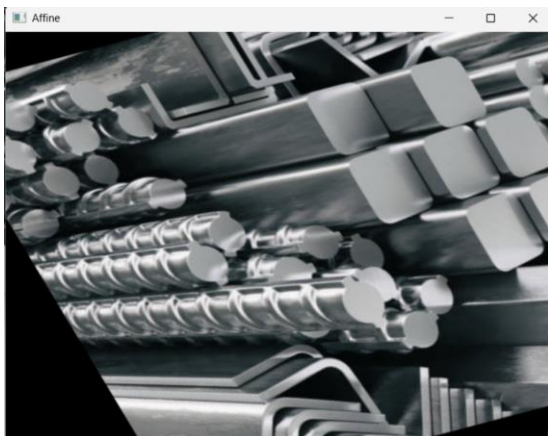
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Sample Input



Sample Output



Result:

The input image was successfully transformed using an affine transformation with the OpenCV library.